

TASK 2

Robotic Arm Simulation

3D CAD Model and Basic Pick-and-Place Demonstration

Software Used: FreeCAD

CAD File: Robotic_Arm_Pick_and_Place.FCStd

1. Introduction

The purpose of this task was to design a basic robotic arm in a 3D CAD environment and demonstrate how it can be used for a simple pick-and-place operation. The model was created in FreeCAD using simple mechanical shapes for the base, arm links, joints, and gripper. The main focus was to understand the basic structure of a robotic arm, its degrees of freedom, and the sequence involved in moving an object from one position to another.

2. Software Used

Software: FreeCAD

Model type: 3D CAD model

File format: FCStd

FreeCAD was used because it is a free and open-source CAD program that provides the tools needed to create and edit basic 3D mechanical models. The final model was saved as a FreeCAD document so that the individual components could be edited later.

3. Robotic Arm Design

The robotic arm was designed as a simplified model rather than a detailed industrial robot. This kept the design easy to understand while still showing the main parts needed for a basic robotic arm. The model contains a circular base, arm links, joint components, and a two-finger gripper. A small cube was also added as the object to be picked and a separate platform was used as the destination.

Main components

Component	Purpose
Base	Supports the arm and represents the base rotation joint.
Shoulder joint	Connects the base to the first arm link.
First arm link	Provides the main reach of the arm.
Elbow joint	Connects the two arm sections and represents elbow movement.
Second arm link	Extends the arm toward the working area.
Gripper	Holds and releases the object.
Object and destination	Used to demonstrate the pick-and-place sequence.

4. Degrees of Freedom (DOF)

For this simplified model, four degrees of freedom were considered. A degree of freedom describes an independent movement that can be controlled in the robotic arm.

DOF	Part	Movement
1	Base	Rotates the arm from side to side.
2	Shoulder	Moves the first arm section.

DOF	Part	Movement
3	Elbow	Moves the second arm section.
4	Gripper	Opens and closes to hold or release an object.

5. Pick-and-Place Operation

A simple pick-and-place sequence was prepared using the CAD model. First, the arm is kept in its initial position with the object placed near the gripper. The arm is then positioned so that the gripper reaches the object. The next position represents the object being held and moved toward the destination. Finally, the object is placed on the destination platform.

Basic sequence: Initial position → Reach the object → Grip the object → Move toward the destination → Place the object → Release the object

Three main CAD states were saved as screenshots: the initial model, the pick position, and the place position. These screenshots are provided separately with the submission.

6. Results

A basic 3D robotic arm model was successfully created in FreeCAD. The model contains the main mechanical sections of a robotic arm and a simple gripper. Four degrees of freedom were identified for the simplified design. The arm was also arranged in separate positions to demonstrate the basic stages of a pick-and-place task.

7. Conclusion

This task provided practical experience with basic 3D CAD modeling and robotic arm design. Using FreeCAD, a simple robotic arm was built from individual components and a four-DOF structure was defined. The pick-and-place example helped demonstrate how the arm can be positioned to reach an object and move it to a different location. Although the model is simplified, it provides a clear starting point for a more advanced robotic arm simulation with real joints, constraints, motors, and dynamic motion.

8. Submission Files

- Robotic_Arm_Pick_and_Place.FCStd — FreeCAD CAD model
- 01_Initial_Model.png — initial model screenshot
- 02_Pick_Position.png — pick position screenshot
- 03_Place_Position.png — place position screenshot
- Task 2 Report.pdf — this report